8086 Microprocessor in Maximum Mode

Q. Explain the operation of the 8086 microprocessor in Maximum Mode. Discuss the role of the Bus Controller (8288) and how a multiprocessor environment is managed.

> **Definition to Remember**

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codes

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1. Maximum Mode Configuration

In this mode, pins that previously generated control signals (like `ALE`, `WR'`) take on new functions related to bus arbitration and multiprocessor coordination.

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2. Reassigned Maximum Mode Signals

- **S2', S1', S0' (Status Lines):** The 8086 outputs its current cycle status on these 3 pins. The 8288 reads them to generate command signals.

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3. Role of the 8288 Bus Controller

The 8288 decodes the `S2'-S0'` signals to generate system controls:

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4. Managing the Multiprocessor Environment

- **Bus Arbitration:** To prevent data corruption, only one processor can use the bus at a time. A coprocessor asserts **RQ'**. The 8086 finishes its cycle, floats its buses (High-Z), and asserts **GT'** to hand over control.

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memo

> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

> `Ma
RQ/GT